large model $\supset$ (nested) small model

$lm(y \sim x_1 + x_2)$ $\quad$ $lm(y \sim x_1)$

$SS_{error}^{large} \leq SS_{error}^{small}$

$p^{large} > p^{small}$

$0 < f = \left(\frac{SS^{small} - SS^{large}}{p^{large} - p^{small}} \right) / (\dots) = 5.6$

If $f \gg 0$, then SS_{error} ↓ faster than p^{par} ↑ ~ "large"

F-Test

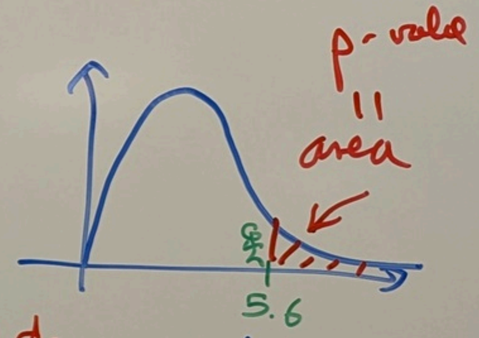
H_0: small model is good enough
no need large model

Under H_0:

$f \sim F_{d_1, d_2}$

$d_1 = p^{large} - p^{small}$

$d_2 = n - p^{large}$



Logic:

large f → small p-value → reject H_0 → need large model

↓

large model is better

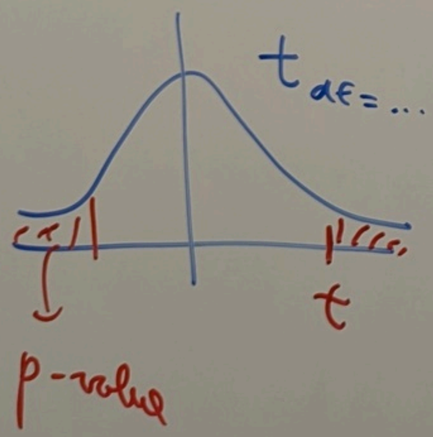
T-test

$x_1, \ldots, x_n \sim N(\mu, \sigma^2)$ — μ: unknown

$H_0: \mu = 0$?

test statistic $t = \dfrac{\bar{x} - 0}{(s.e.)}$ ← 10^{-10} ; 0.0001

Reject



$H_0: \beta_2 = 0$?

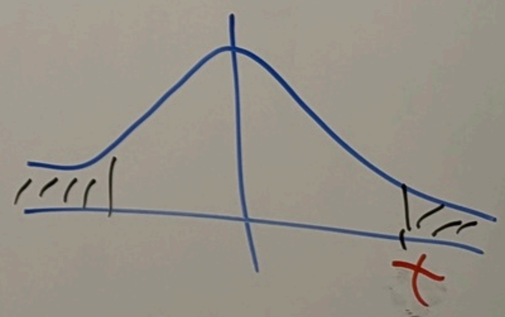
$t = \dfrac{\hat{\beta}_2 - 0}{(s.e.)_?}$

$\hat{\beta}_2 \pm$ s.e.

-0.03 ± 0.18 ← has 0

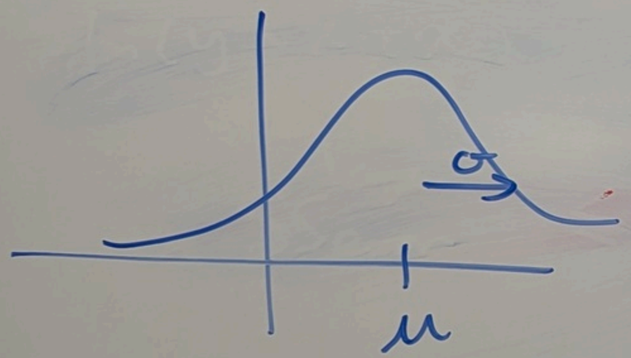
Under H_0: $\beta_2 = 0$ i.e. $lm(y \sim x_1 + x_2)$?? $lm(y \sim x_1)$

$t \sim t^{dist}_{df = n-p}$

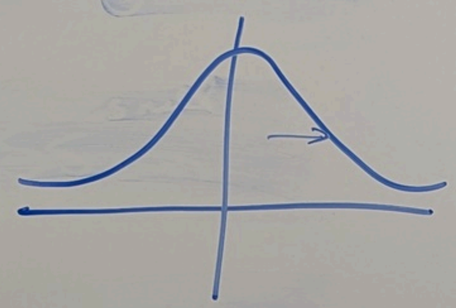


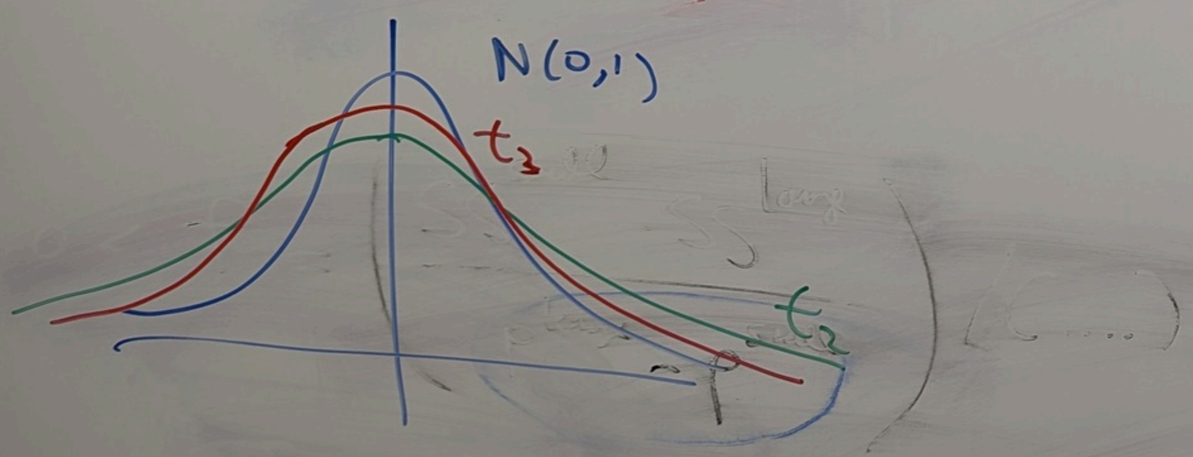
Logic: t large → small p-value → reject H_0 → need large $\beta_2 \neq 0$ → need large $lm(y \sim x_1 + x_2)$

$N(\mu, \sigma^2)$



$t_{df=\nu}$





$t_{df=\nu \to \infty} \approx N(0,1)$

11, 20-30

$\hat{\beta}_0$

$\hat{\beta}_0$ s.e.

0.58 ± 0.34

↑ plausible: $\beta_0 = 0$

t-test. $H_0: \beta_0 = 0$

large p-value 0.51

cannot reject H_0

i.e. $\beta_0 = 0$

↓

is useless predictor

"$y = \beta_2 x_2 + \varepsilon$"?

large model

lm(y~ .... + x_5)

small model

lm(y~ ....)

H_0: small model is enough

$\beta_5 = 0$

F-test

anova(sm, Lm)

T-test

summary(model)

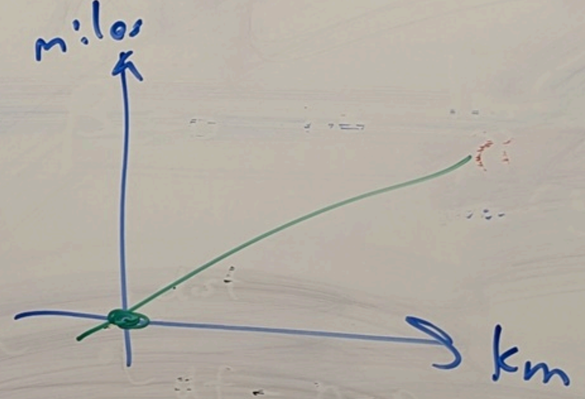
p(|t|)

x_5

p-value ? ?

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$

$y = 0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ ×



Back 7:48 pm

non-nested model

$lm(y \sim x_1)$ vs $lm(y \sim x_2 + x_3)$

ML:
① fit both models
② eval on test set

$\log(1000) = 3$
$\log(10^6)$ 6

model (goodness of fit) $+ \lambda$ (complexity)

min AIC: $-2\log L + 2p$ $+(\log n)$ $p+1$

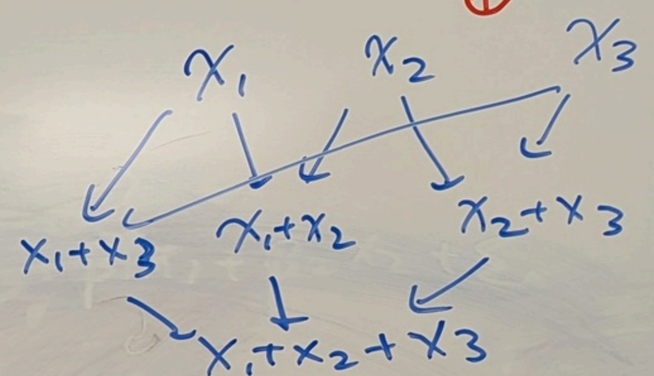
min BIC: $-2\log L + (\log n)\,p$

larger model

$L \uparrow$
$\log L \uparrow$
$-2\log L \downarrow$

$p \uparrow$

FWD ↓



BWD ↑

$lm(y \sim x_1)$ $lm(y \sim x_2 + x_3)$

$lm(y \sim x_1 + x_2 + x_3)$ higher R^2 ↔ ↓ ~~SS~~error

$x_1, x_2, \ldots x_{1000}$

$y \sim \ldots$!

Fwd: $y \sim 1$ AIC 10.5

F-test → $y \sim x_2$ 9.8 ← best

→ $y \sim x_2 + x_5$ 12.7

⋮

"LASSO"

SS_{error}

$\frac{}{n-p}$

Bwd: $y \sim x_1 + \ldots + x_{1000}$

$y \sim x_1 + \ldots + x_{999}$ ← t-test remove highest p-value until all < 0.

⋮

stop

y = money

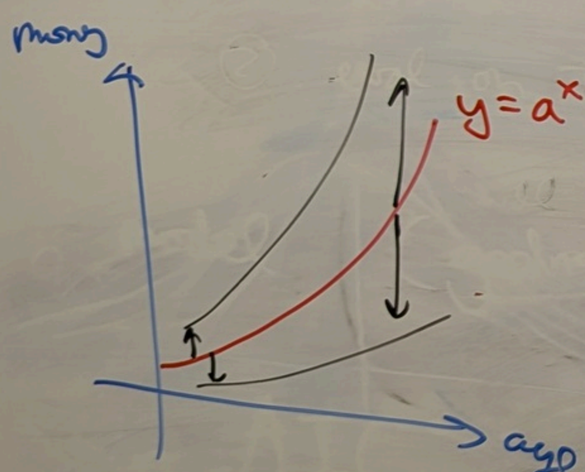
x = age

$$MSE_{loss} = \frac{1}{n}\sum_{i=1}^{\hat{n}} (y_i - \hat{y}_i)^2 \cdot \underline{w_i}$$

1st model: $lm(y \sim x)$

$$(money) = \beta_0 + \beta_1 (age) + \varepsilon \quad \leftarrow N(0, \sigma^2)$$

ε: $Var(\varepsilon) = \sigma^2$



$$y = \underbrace{\beta_0 \cdot \beta_1^{(x)}}_{\text{multiplicative factor}} \cdot \varepsilon \quad \leftarrow Var(\varepsilon) = const$$

noise in scale %

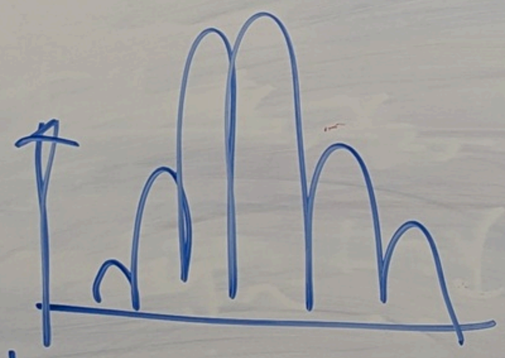
$lm(\log(y) \sim x)$

$lm(y \sim e^x)$?

model ← lm(y ~ ...)

residuals $r_1, r_2, \ldots, r_n \sim N(\ldots, \ldots)$?

data $r_1 \le r_2 \le \cdots \le r_{100}$



hist($\vec{r}$)

